

Problem A. Approximate Answer

Time Limit 1000 ms

Code Length Limit 50000 B

OS Linux

You are solving a problem whose correct answer is the integer Y .

The answer you obtained, however, is the integer X .

Your answer will be considered correct if the difference between X and Y is at most K .

In other words, your answer is considered correct if and only if:

$$|X - Y| \leq K$$

Given the values of X , Y , and K , determine whether your answer X is correct or not.

Note that $|x|$ denotes the [absolute value](#) of x .

For example, $|12| = 12$, $|-15| = 15$, and $|0| = 0$.

Input Format

- The first and only line of input will contain three space-separated integers X , Y and K — your answer, the correct answer, and the the maximum allowed difference, respectively.

Output Format

Output the answer on a single line: **"Yes"** if your answer is considered correct, and **"No"** otherwise (without quotes).

Each letter of the output may be printed in either uppercase or lowercase, i.e, the strings

NO, **no**, **No**, and **n0** will all be treated as equivalent.

Constraints

- $1 \leq X, Y, K \leq 20$

Sample 1

Input	Output
10 5 4	No

$|X - Y| = |10 - 5| = 5$, while $K = 4$. Since $|X - Y| > K$, the answer is not considered correct.

Sample 2

Input	Output
10 5 5	Yes

$|X - Y| = |10 - 5| = 5$, while $K = 5$. Since $|X - Y| \leq K$, the answer is considered correct.

Problem B. Buttons

Time Limit 2000 ms

Problem Statement

There are two buttons, one of size A and one of size B .

When you press a button of size X , you get X coins and the size of that button decreases by 1.

You will press a button twice. Here, you can press the same button twice, or press both buttons once.

At most how many coins can you get?

Constraints

- All values in input are integers.
- $3 \leq A, B \leq 20$

Input

Input is given from Standard Input in the following format:

A B

Output

Print the maximum number of coins you can get.

Sample 1

Input	Output
5 3	9

You can get $5 + 4 = 9$ coins by pressing the button of size 5 twice, and this is the maximum result.

Sample 2

Input	Output
3 4	7

Sample 3

Input	Output
6 6	12

Problem C. Not Found

Time Limit 2000 ms

Problem Statement

You are given a string S consisting of lowercase English letters. Find the lexicographically (alphabetically) smallest lowercase English letter that does not occur in S . If every lowercase English letter occurs in S , print **None** instead.

Constraints

- $1 \leq |S| \leq 10^5$ ($|S|$ is the length of string S .)
- S consists of lowercase English letters.

Input

Input is given from Standard Input in the following format:

S

Output

Print the lexicographically smallest lowercase English letter that does not occur in S . If every lowercase English letter occurs in S , print **None** instead.

Sample 1

Input	Output
atcoderregularcontest	b

The string `atcoderregularcontest` contains **a**, but does not contain **b**.

Sample 2

Input	Output
abcdefghijklmnopqrstuvwxyz	None

This string contains every lowercase English letter.

Sample 3

Input	Output
fajsonlslfepbjtsaayxbymeskptcumtwrmkkinjx nnucagfrg	d

Problem D. How many?

Time Limit 2000 ms

Problem Statement

How many triples of non-negative integers (a, b, c) satisfy $a + b + c \leq S$ and $a \times b \times c \leq T$?

Constraints

- $0 \leq S \leq 100$
- $0 \leq T \leq 10000$
- S and T are integers.

Input

Input is given from Standard Input in the following format:

S T

Output

Print the number of triples of non-negative integers (a, b, c) satisfying the conditions.

Sample 1

Input	Output
1 0	4

The triples (a, b, c) satisfying the conditions are $(0, 0, 0)$, $(0, 0, 1)$, $(0, 1, 0)$, and $(1, 0, 0)$ — there are four of them.

Sample 2

Input	Output
2 5	10

Sample 3

Input	Output
10 10	213

Sample 4

Input	Output
30 100	2471

Problem E. Minor Change

Time Limit 2000 ms

Problem Statement

Given are strings S and T . Consider changing S to T by repeating the operation below. Find the minimum number of operations required to do so.

Operation: Choose one character of S and replace it with a different character.

Constraints

- S and T have lengths between 1 and 2×10^5 (inclusive).
- S and T consists of lowercase English letters.
- S and T have equal lengths.

Input

Input is given from Standard Input in the following format:

S
 T

Output

Print the answer.

Sample 1

Input	Output
cupofcoffee cupofhottea	4

We can achieve the objective in four operations, such as the following:

- First, replace the sixth character **c** with **h**.
- Second, replace the eighth character **f** with **t**.

- Third, replace the ninth character **f** with **t**.
- Fourth, replace the eleventh character **e** with **a**.

Sample 2

Input	Output
abcde bcdea	5

Sample 3

Input	Output
apple apple	0

No operations may be needed to achieve the objective.

Problem F. Find Multiple

Time Limit 2000 ms

Problem Statement

Print a number between A and B (inclusive) that is a multiple of C .

If there is no such number, print **-1**.

Constraints

- $1 \leq A \leq B \leq 1000$
- $1 \leq C \leq 1000$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

A B C

Output

Print the answer.

If there is no number with the desired property, print **-1**.

Sample 1

Input	Output
123 456 100	200

300 or **400** would also be accepted.

Sample 2

Input	Output
630 940 314	-1

Problem G. Biscuit Generator

Time Limit 2000 ms

Problem Statement

A biscuit making machine produces B biscuits at the following moments: A seconds, $2A$ seconds, $3A$ seconds and each subsequent multiple of A seconds after activation.

Find the total number of biscuits produced within $T + 0.5$ seconds after activation.

Constraints

- All values in input are integers.
- $1 \leq A, B, T \leq 20$

Input

Input is given from Standard Input in the following format:

$A \ B \ T$

Output

Print the total number of biscuits produced within $T + 0.5$ seconds after activation.

Sample 1

Input	Output
3 5 7	10

- Five biscuits will be produced three seconds after activation.
- Another five biscuits will be produced six seconds after activation.
- Thus, a total of ten biscuits will be produced within 7.5 seconds after activation.

Sample 2

Input	Output
3 2 9	6

Sample 3

Input	Output
20 20 19	0

Problem H. Missing Number

Time Limit 1000 ms

Mem Limit 524288 kB

You are given all numbers between $1, 2, \dots, n$ except one. Your task is to find the missing number.

Input

The first input line contains an integer n .

The second line contains $n - 1$ numbers. Each number is distinct and between 1 and n (inclusive).

Output

Print the missing number.

Constraints

- $2 \leq n \leq 2 \cdot 10^5$

Example

Input	Output
5 2 3 1 5	4

Problem I. Increasing Array

Time Limit 1000 ms

Mem Limit 524288 kB

You are given an array of n integers. You want to modify the array so that it is increasing, i.e., every element is at least as large as the previous element.

On each move, you may increase the value of any element by one. What is the minimum number of moves required?

Input

The first input line contains an integer n : the size of the array.

Then, the second line contains n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print the minimum number of moves.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
5 3 2 5 1 7	5

Problem J. Gaming Forces

Time Limit 2000 ms

Mem Limit 262144 kB

Monocarp is playing a computer game. He's going to kill n monsters, the i -th of them has h_i health.

Monocarp's character has two spells, either of which he can cast an arbitrary number of times (possibly, zero) and in an arbitrary order:

- choose exactly two alive monsters and decrease their health by 1;
- choose a single monster and kill it.

When a monster's health becomes 0, it dies.

What's the minimum number of spell casts Monocarp should perform in order to kill all monsters?

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of testcases.

The first line of each testcase contains a single integer n ($1 \leq n \leq 100$) — the number of monsters.

The second line contains n integers $h_1, h_2, \dots, h_n$ ($1 \leq h_i \leq 100$) — the health of each monster.

The sum of n over all testcases doesn't exceed $2 \cdot 10^4$.

Output

For each testcase, print a single integer — the minimum number of spell casts Monocarp should perform in order to kill all monsters.

Examples

Input	Output
3 4 1 2 1 2 3 2 4 2 5 1 2 3 4 5	3 3 5

Note

In the first testcase, the initial health list is $[1, 2, 1, 2]$. Three spells are casted:

- the first spell on monsters 1 and 2 — monster 1 dies, monster 2 has now health 1, new health list is $[0, 1, 1, 2]$;
- the first spell on monsters 3 and 4 — monster 3 dies, monster 4 has now health 1, new health list is $[0, 1, 0, 1]$;
- the first spell on monsters 2 and 4 — both monsters 2 and 4 die.

In the second testcase, the initial health list is $[2, 4, 2]$. Three spells are casted:

- the first spell on monsters 1 and 3 — both monsters have health 1 now, new health list is $[1, 4, 1]$;
- the second spell on monster 2 — monster 2 dies, new health list is $[1, 0, 1]$;
- the first spell on monsters 1 and 3 — both monsters 1 and 3 die.

In the third testcase, the initial health list is $[1, 2, 3, 4, 5]$. Five spells are casted. The i -th of them kills the i -th monster with the second spell. Health list sequence: $[1, 2, 3, 4, 5] \rightarrow [0, 2, 3, 4, 5] \rightarrow [0, 0, 3, 4, 5] \rightarrow [0, 0, 0, 4, 5] \rightarrow [0, 0, 0, 0, 5] \rightarrow [0, 0, 0, 0, 0]$.

Problem K. Mahmoud and Ehab and the MEX

Time Limit 2000 ms

Mem Limit 262144 kB

Dr. Evil kidnapped Mahmoud and Ehab in the evil land because of their performance in the Evil Olympiad in Informatics (EOI). He decided to give them some problems to let them go.

Dr. Evil is interested in sets, He has a set of n integers. Dr. Evil calls a set of integers *evil* if the *MEX* of it is exactly x . the *MEX* of a set of integers is the minimum non-negative integer that doesn't exist in it. For example, the *MEX* of the set $\{0, 2, 4\}$ is 1 and the *MEX* of the set $\{1, 2, 3\}$ is 0.

Dr. Evil is going to make his set *evil*. To do this he can perform some operations. During each operation he can add some non-negative integer to his set or erase some element from it. What is the minimal number of operations Dr. Evil has to perform to make his set *evil*?

Input

The first line contains two integers n and x ($1 \leq n \leq 100$, $0 \leq x \leq 100$) — the size of the set Dr. Evil owns, and the desired *MEX*.

The second line contains n distinct non-negative integers not exceeding 100 that represent the set.

Output

The only line should contain one integer — the minimal number of operations Dr. Evil should perform.

Examples

Input	Output
5 3 0 4 5 6 7	2

Input	Output
1 0 0	1

Input	Output
5 0 1 2 3 4 5	0

Note

For the first test case Dr. Evil should add 1 and 2 to the set performing 2 operations.

For the second test case Dr. Evil should erase 0 from the set. After that, the set becomes empty, so the *MEX* of it is 0.

In the third test case the set is already *evil*.

Problem L. Twin Permutations

Time Limit 1000 ms

Mem Limit 262144 kB

You are given a permutation[†] a of length n .

Find any permutation b of length n such that $a_1 + b_1 \leq a_2 + b_2 \leq a_3 + b_3 \leq \dots \leq a_n + b_n$.

It can be proven that a permutation b that satisfies the condition above always exists.

[†] A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, $[2, 3, 1, 5, 4]$ is a permutation, but $[1, 2, 2]$ is not a permutation (2 appears twice in the array), and $[1, 3, 4]$ is also not a permutation ($n = 3$ but there is 4 in the array).

Input

Each test contains multiple test cases. The first line of input contains a single integer t ($1 \leq t \leq 2000$) — the number of test cases. The description of test cases follows.

The first line of each test case contains a single integer n ($1 \leq n \leq 100$) — the length of permutations a and b .

The second line of each test case contains n distinct integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq n$) — the elements of permutation a . All elements of a are distinct.

Note that there is no bound on the sum of n over all test cases.

Output

For each test case, output **any** permutation b which satisfies the constraints mentioned in the statement. It can be proven that a permutation b that satisfies the condition above always exists.

Examples

Input	Output
5 5 1 2 4 5 3 2 1 2 1 1 3 3 2 1 4 1 4 3 2	1 2 4 3 5 2 1 1 1 2 3 1 2 3 4

Note

In the first test case $a = [1, 2, 4, 5, 3]$. Then the permutation $b = [1, 2, 4, 3, 5]$ satisfies the condition because $1 + 1 \leq 2 + 2 \leq 4 + 4 \leq 5 + 3 \leq 3 + 5$.

Problem M. Cypher

Time Limit 1000 ms

Mem Limit 262144 kB

Luca has a cypher made up of a sequence of n wheels, each with a digit a_i written on it. On the i -th wheel, he made b_i moves. Each move is one of two types:

- *up* move (denoted by U): it increases the i -th digit by 1. After applying the up move on 9, it becomes 0.
- *down* move (denoted by D): it decreases the i -th digit by 1. After applying the down move on 0, it becomes 9.



Example for $n = 4$. The current sequence is 0 0 0 0.

Luca knows the final sequence of wheels and the moves for each wheel. Help him find the original sequence and crack the cypher.

Input

The first line contains a single integer t ($1 \leq t \leq 100$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \leq n \leq 100$) — the number of wheels.

The second line contains n integers a_i ($0 \leq a_i \leq 9$) — the digit shown on the i -th wheel after all moves have been performed.

Then n lines follow, the i -th of which contains the integer b_i ($1 \leq b_i \leq 10$) and b_i characters that are either U or D — the number of moves performed on the i -th wheel, and the moves

performed. U and D represent an *up* move and a *down* move respectively.

Output

For each test case, output n space-separated digits — the initial sequence of the cypher.

Examples

Input	Output
3	2 1 1
3	9 0
9 3 1	0 4 9 6 9
3 DDD	
4 UDUU	
2 DU	
2	
0 9	
9 DDDDDDDDD	
9 UUUUUUUUU	
5	
0 5 9 8 3	
10 UUUUUUUUUU	
3 UUD	
8 UUDUDDDD	
10 UUDUUDUDDU	
4 UUUU	

Note

In the first test case, we can prove that initial sequence was $[2, 1, 1]$. In that case, the following moves were performed:

- On the first wheel: $2 \xrightarrow{D} 1 \xrightarrow{D} 0 \xrightarrow{D} 9$.
- On the second wheel: $1 \xrightarrow{U} 2 \xrightarrow{D} 1 \xrightarrow{U} 2 \xrightarrow{U} 3$.
- On the third wheel: $1 \xrightarrow{D} 0 \xrightarrow{U} 1$.

The final sequence was $[9, 3, 1]$, which matches the input.

Problem N. Chemistry

Time Limit 2000 ms

Mem Limit 262144 kB

You are given a string s of length n , consisting of lowercase Latin letters, and an integer k .

You need to check if it is possible to remove **exactly** k characters from the string s in such a way that the remaining characters can be rearranged to form a palindrome. Note that you can reorder the remaining characters in any way.

A palindrome is a string that reads the same forwards and backwards. For example, the strings "z", "aaa", "aba", "abccba" are palindromes, while the strings "codeforces", "reality", "ab" are not.

Input

Each test consists of multiple test cases. The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of the test cases. This is followed by their description.

The first line of each test case contains two integers n and k ($0 \leq k < n \leq 10^5$) — the length of the string s and the number of characters to be deleted.

The second line of each test case contains a string s of length n , consisting of lowercase Latin letters.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output "YES" if it is possible to remove exactly k characters from the string s in such a way that the remaining characters can be rearranged to form a palindrome, and "NO" otherwise.

You can output the answer in any case (uppercase or lowercase). For example, the strings "yEs", "yes", "Yes", and "YES" will be recognized as positive answers.

Examples

Input	Output
14	YES
1 0	NO
a	YES
2 0	YES
ab	YES
2 1	YES
ba	NO
3 1	NO
abb	YES
3 2	YES
abc	YES
6 2	YES
bacacd	NO
6 2	YES
fagbza	
6 2	
zwaafa	
7 2	
taagaak	
14 3	
tttraakkttoorr	
5 3	
debdb	
5 4	
ecadc	
5 3	
debca	
5 3	
abaac	

Note

In the first test case, nothing can be removed, and the string "a" is a palindrome.

In the second test case, nothing can be removed, but the strings "ab" and "ba" are not palindromes.

In the third test case, any character can be removed, and the resulting string will be a palindrome.

In the fourth test case, one occurrence of the character "a" can be removed, resulting in the string "bb", which is a palindrome.

In the sixth test case, one occurrence of the characters "b" and "d" can be removed, resulting in the string "acac", which can be rearranged to the string "acca".

In the ninth test case, one occurrence of the characters "t" and "k" can be removed, resulting in the string "aagaa", which is a palindrome.